

AI-Powered Driver Drowsiness Detection System

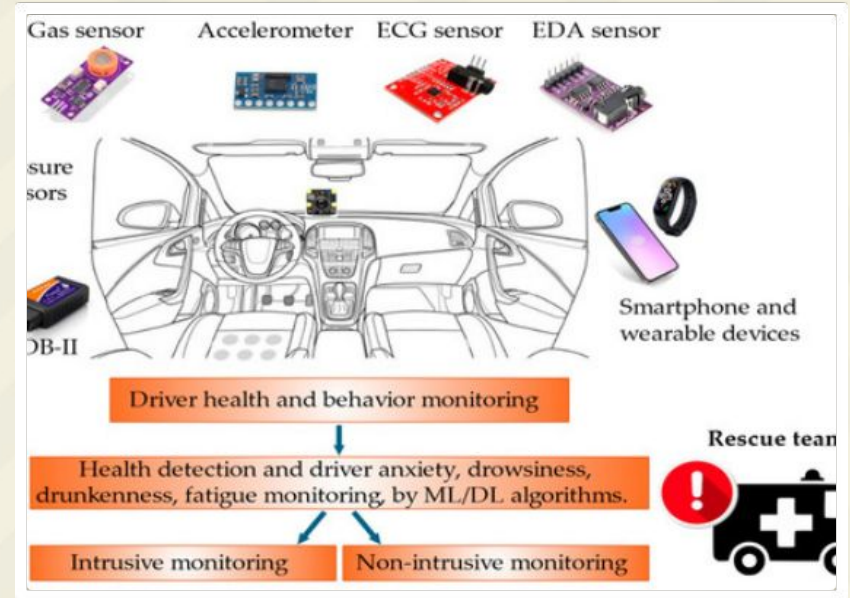
Milestone 1 Presentation

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PROBLEM & MOTIVATION

Safety Crisis on Roads

- **Night & Long-haul Safety:** Driver fatigue is a leading cause of fatal accidents in commercial transport.
- **Urbanization Trends:** Increasing cab and private vehicle usage demands automated alertness monitoring.
- **Proactive Warnings:** The system aims to provide an additional safety layer before fatigue becomes dangerous.



SCOPE & STAKEHOLDERS



Hardware Scope

Prioritizing

Dashboard/Embedded

Cameras over smartphones to avoid battery drain and multitasking interference.



Stakeholders

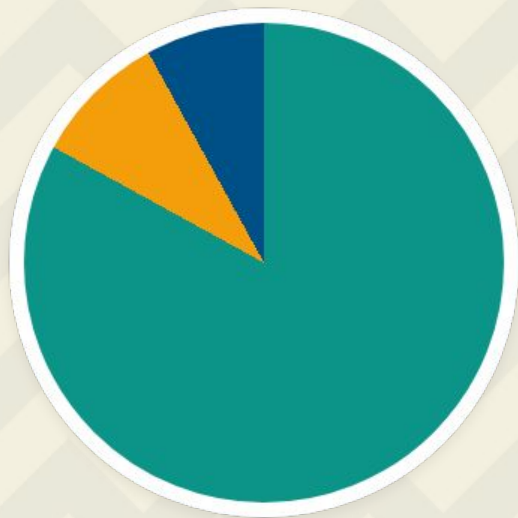
Logistics agencies, cab owners, bus operators, and individual drivers aiming for highway safety parameters.



Exclusions

Out of Scope: Real-car mechanical braking integration, mobile application development, and cloud dashboards.

DATASET UNDERSTANDING



Roboflow Driver Dataset

21,668 Total Labeled Images

- Training Set: 17,961 (83%)
- Validation Set: 1,881 (9%)
- Testing Set: 1,826 (8%)

CLASSES & HORIZON



Class Distribution

- **Open Eyes:** 9,510 objects
- **Yawn States:** 7,196 objects
- **Closed Eyes:** 6,804 objects

Next Steps: Temporal Data

Integrating video frameworks for sequence validation in Milestone 2:

- **NTHU Dataset:** For sequence tracking and varied lighting profiles.
- **YawDD Dataset:** For specialized continuous yawning validation.



RULE-BASED: MEDIAPIPE

Facial Landmark Logic

- **EAR (Eye Aspect Ratio):** Calculating distances between vertical eyelid points to isolate blinks and closures.
- **MAR (Mouth Aspect Ratio):** Monitoring vertical lip separation to identify distinct yawning trends.
- **OpenCV:** Real-time frame processing and handling runtime threshold overlays.



PERCLOS & POSE LOGIC



PERCLOS

Tracking the percentage of eye closure over time to filter out standard, rapid blinks from genuine fatigue.



Head Pose

Detecting nodding or slumping events by measuring head tilt (pitch/yaw) indicating fatigue.



Limitations

Fixed ratios lose accuracy during lighting shifts, camera angle offsets, or when drivers wear eyewear.

DEEP LEARNING: YOLO



Robust State Identification

- **Direct Classification:** Detecting Closed Eye or Yawn boxes directly from raw pixels.
- **Resiliency:** Convolutional weights adapt better to cabin noise, obstructions, and low lighting.
- **Models:** Utilizing **YOLOv8n** or **YOLO11n** for high FPS on dashboard hardware.

PROPOSED HYBRID PIPELINE

Input Feed

MediaPipe

YOLO Analysis

Sliding Window

Staged Alert

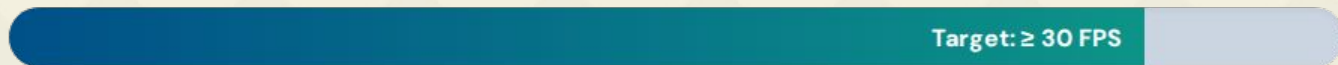
Proposed Paradigm: Landmark precision cross-checked against YOLO detections, smoothed by temporal sliding windows for reliable, real-time alerts.

EVALUATION METRICS

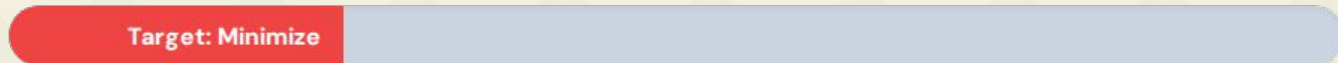
Precision & Recall



Processing Speed



False Alarm Rate



Testing via **mAP** layers and tracking stability over time using balanced **F1-scores**.

TESTING & ALERTS



Environments

Validating across lighting challenges, tunnel simulations, eyewear variations, and deep angle changes.



Alert Logic

Stage 1: Alert State (Green)

Stage 2: Mild Fatigue (Orange Visual)

Stage 3: Drowsy Warning (Red Audio)



Expected Output

Continuous fatigue score log records paired with immediate onscreen tracking overlays.

REFERENCES

Roboflow Driver Drowsiness Public Dataset Engine Hub (2024)

MediaPipe Face Mesh Processing Framework – Google Engineering

Ultralytics YOLO Framework Architecture Repository

Soukupova & Cech: Real-Time Eye Blink Detection (EAR Reference)

Thank You!

Questions? | Milestone 1 Final Presentation